Change-Aware Siamese U-Net for Satellite-Based Post-Disaster Building Damage Segmentation with Grad-CAM Explainability

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Abstract— Natural disasters including earthquakes, floods, and hurricanes often result in widespread structural damage, creating the need for fast and scalable assessment methods. This paper proposes an AI-driven post-disaster damage assessment framework that processes paired pre-disaster and post-disaster satellite imagery using a Siamese U-Net with change-aware temporal fusion. The network extracts features from pre-event and post-event image tiles through a shared encoder and combines temporal information at every scale by concatenating post-event features with the absolute difference between pre-event and post-event feature representations. Trained on an xBD-style dataset containing 87 paired tiles across five damage categories, the proposed framework achieves an overall pixel accuracy of 88.27% and an AUC of 0.8435, while obtaining a Background class Dice score of 0.9428. Grad-CAM and Score-CAM explainability modules generate spatially aligned saliency maps that improve the interpretability of model predictions. An operational Streamlit dashboard enables tile-level inference together with ground truth comparison, supporting practical deployment in real-world disaster assessment.

Keywords: Disaster Assessment, Satellite Imagery, Siamese U-Net, Semantic Segmentation, Grad-CAM, xBD, Change Detection

I. INTRODUCTION

Over recent years, the frequency of natural disasters such as earthquakes, floods, hurricanes, and wildfires has increased considerably across many parts of the world. These events often lead to extensive loss of human life, severe damage to infrastructure, and significant economic disruption. Therefore, the rapid and accurate assessment of affected regions is essential for supporting emergency response, ensuring efficient resource allocation, and guiding both immediate recovery and long-term rehabilitation efforts. Conventional damage assessment methods mainly depend on manual inspections, making the process slow, labor-intensive, and frequently delayed due to limited accessibility in disaster-stricken locations.

Recent advances in remote sensing have enabled the availability of high-resolution satellite imagery, making it possible to observe extensive geographical regions in near real time. At the same time, progress in artificial intelligence, particularly deep learning, has made it possible to automatically analyze complex visual information with high efficiency. The combination of these technologies offers a powerful and scalable approach for conducting post-disaster damage assessment over large affected areas. Recent advances in spaceborne imaging platforms have enabled near-continuous coverage of disaster-affected regions at meter-scale resolution, while convolutional architectures have matured to the point where pixel-level structural analysis from overhead imagery is operationally tractable. This work brings these capabilities together in a unified pipeline that pairs a change-aware Siamese encoder with a multi-scale decoder, trained under a class-imbalance-aware objective on xBD-style annotated tiles. To support operational adoption, the system exposes predictions through an interactive Streamlit interface and provides spatially registered saliency maps via Grad-CAM and Score-CAM, enabling analysts to verify which image regions drove each damage classification.

The key contributions of this work are:

1. A Siamese U-Net with per-scale temporal difference fusion (concatenation of post features and |post - pre| features) for 5-class building damage segmentation.

2. A combined Soft Dice and weighted Cross-Entropy loss to address extreme class imbalance across five damage severity categories.

3. Integrated Grad-CAM and Score-CAM explainability targeting the final decoder convolution for spatially aligned saliency maps.

4. An operational Streamlit dashboard supporting tile-level inference on xBD-format satellite image pairs with ground truth comparison.

II. LITERATURE SURVEY

Recent advancements in deep learning and remote sensing have significantly improved post-disaster damage assessment using satellite imagery. Researchers have increasingly focused on developing automated systems capable of

identifying damaged buildings, classifying damage severity, and assisting emergency responders with rapid decision-making.

[1] Alisjahbana et al. addressed the challenge of building-level damage mapping by decomposing the problem into sequential localization and severity estimation stages, using bi-temporal satellite imagery as input. While their approach improved upon the xView2 competition baseline, it operates exclusively on visual data and does not account for auxiliary contextual signals such as terrain attributes or meteorological conditions that may influence damage patterns across different disaster types.

[2] Xue et al. demonstrated that incorporating structured property metadata alongside visual inputs can improve hurricane damage classification beyond what image-only models achieve. Their Swin Transformer-based architecture fuses street-level photographs with tabular fields including construction age and recorded wind exposure. The framework is tailored to a single disaster type and depends on ground-level imagery, constraining its applicability to wide-area satellite-based monitoring pipelines.

[3] Xiao and Mostafavi proposed a hierarchical transformer approach that moves beyond binary damaged/undamaged labelling toward finer structural typologies such as roof integrity and wall failure. The architecture reports strong segmentation quality metrics on post-event imagery, providing richer outputs for operational response planning. However, the system does not incorporate meteorological or geospatial auxiliary data and provides no mechanism for visual explanation of individual predictions.

[4] Gebre et al. explored cross-attention mechanisms operating over pairs of bi-temporal satellite tiles to produce multi-class building condition labels, with additional robustness to partially missing observations. Despite strong classification results, the framework is oriented toward feature-level fusion and does not generate spatially interpretable outputs that would allow an operator to verify which image regions drove a given prediction.

The reviewed studies demonstrate that deep learning has substantially advanced automated disaster damage assessment. However, most existing approaches either rely solely on satellite imagery or address segmentation and classification independently. Very few methods integrate change-aware temporal fusion, class-imbalance-robust training objectives, explainable saliency visualization, and an operational inference interface within a single unified framework. These gaps motivate the present work, which combines Siamese encoder-based temporal difference fusion, combined Dice and weighted Cross-Entropy loss, custom Grad-CAM and Score-CAM explainability, and a Streamlit dashboard for end-to-end post-disaster damage assessment.

III. Methodology

This section describes the end-to-end pipeline for post-disaster damage segmentation, covering dataset preparation, preprocessing, model architecture, loss formulation, and explainability. The overall system flow is illustrated in Fig. 1.

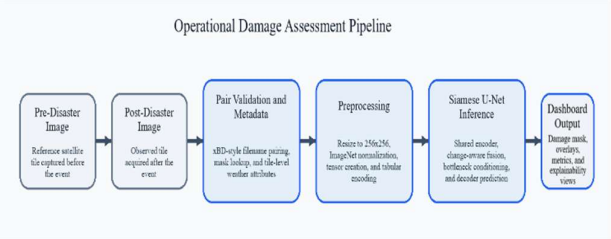


Fig 1. Operational damage assessment pipeline showing end-to-end flow from pre/post satellite tile pair to dashboard output.

A. Dataset and Pairing

The PRJ-3563 dataset comprises 87 paired satellite tiles in xBD format. Image filenames follow the pattern <tile_id>_<pre|post>_disaster.<ext>, enabling deterministic pairing of pre-event and post-event tiles by tile identifier. The training target is the post-disaster damage mask, which assigns each pixel to one of five classes: Background (0), No Damage (1), Minor Damage (2), Major Damage (3), and Destroyed (4). The dataset is split into 69 training tiles and 18 validation tiles. Class distribution is severely imbalanced, with Background and No Damage pixels dominating and Destroyed pixels being extremely rare.

Fig. 2 presents three representative tile pairs from the validation split, illustrating the visual diversity of pre/post imagery and the corresponding ground truth damage masks. The color overlay confirms the dominance of No Damage (green) and Background pixels across tiles, with Minor and Major Damage appearing at the building level and Destroyed pixels being visually sparse even in heavily affected scenes.

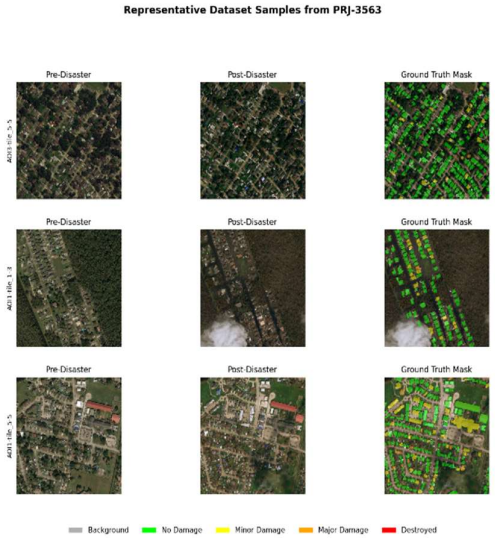


Fig 2. Representative pre/post image pairs and ground truth damage masks from the PRJ-3563 validation split. Overlay colors: green=no damage, yellow=minor damage, orange=major damage, red=destroyed.

B. Preprocessing

Input tiles are spatially resampled to a fixed 256×256 pixel resolution prior to network ingestion. Channel-wise intensity

normalization follows the population statistics of the ImageNet corpus [20], a standard practice when initializing encoders with weights pretrained on large-scale visual recognition tasks. The normalized pre-event and post-event tensors are then stacked along the channel axis to produce a six-channel representation that serves as the unified input to the shared encoder.

C. Model Architecture - SiameseUNet

The proposed model is a Siamese U-Net in which a shared-weight encoder processes the pre-event and post-event image branches independently. At each of four encoder scales, temporal information is fused by concatenating the post-event feature map with the absolute difference between post-event and pre-event feature maps: fusion = concat(post_features, |post_features - pre_features|). Four decoder blocks with skip connections progressively recover spatial resolution, and a final 1x1 convolution produces a 5-class logit map. The architecture additionally supports optional FiLM-style tabular conditioning at the bottleneck for meteorological inputs (rainfall, wind speed, elevation, slope); this pathway was not activated in the current evaluation due to dataset constraints. The architecture is shown in Fig. 3.

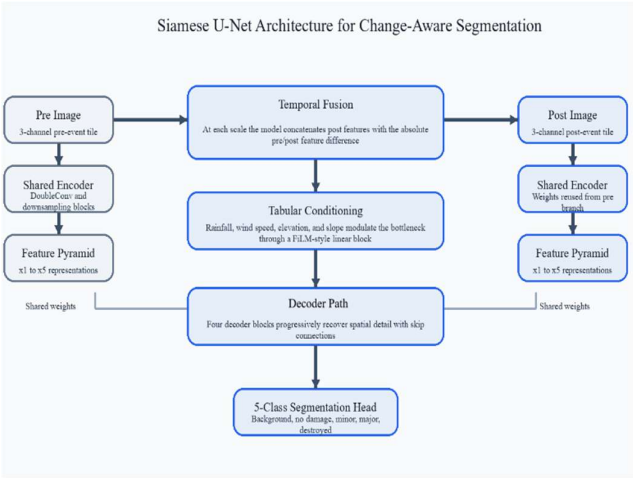


Fig 3. Siamese U-Net architecture with shared encoder, per-scale temporal difference fusion, and 5-class segmentation head.

D. Loss Function

The training objective is a linear combination of SoftDiceLoss and weighted CrossEntropyLoss, motivated by the complementary properties of the two terms [15]. CrossEntropyLoss penalizes per-pixel misclassification but is sensitive to class frequency, assigning disproportionately low gradients to rare categories. SoftDiceLoss directly optimizes the overlap between predicted and ground truth regions, providing a frequency-independent signal that benefits minority classes. Class weights for the CrossEntropy term are estimated from a full pixel-frequency scan of all 69 training masks, producing inverse-frequency weights that amplify gradient contributions from the Minor Damage, Major Damage, and Destroyed categories.

E. Explainability

A custom Grad-CAM implementation targets the final decoder convolution layer to produce class-discriminative saliency maps. Score-CAM is implemented as a gradient-free alternative. Both maps are normalized and upsampled to match the input resolution before being overlaid on the post-disaster image for spatial alignment, as shown in Fig. 4.

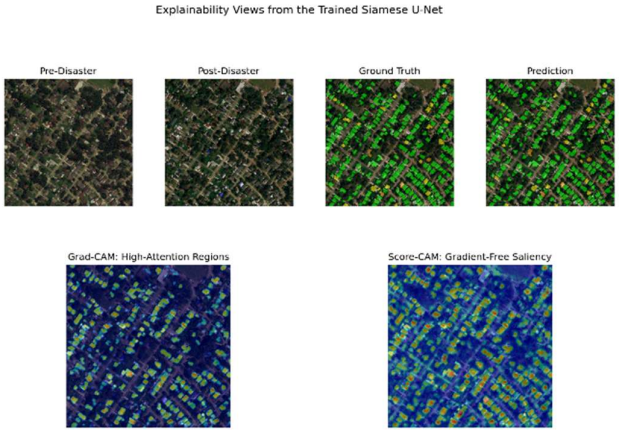


Fig 4. Explainability views from the trained Siamese U-Net: prediction panel (top) and Grad-CAM/Score-CAM saliency overlays (bottom).

F. Operational Dashboard

An operational Streamlit dashboard supports tile-level inference on xBD-format image pairs. Users can select dataset tiles or upload custom image pairs, view predicted damage masks, compare against ground truth annotations, and inspect Grad-CAM activation overlays. The dashboard is shown in Fig. 5.

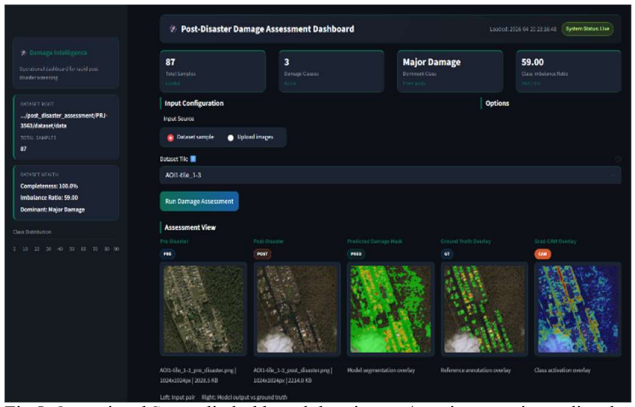


Fig 5. Operational Streamlit dashboard showing pre/post image pair, predicted damage mask, ground truth overlay, and Grad-CAM activation.

IV. Experiments

A. Training Setup

Two training runs were conducted using the Adam optimizer with a learning rate of 1e-3 and weight decay of 1e-4, batch size 2, on the 69-tile training split. Data augmentation was enabled for the full training runs and disabled for baseline overfit testing.

The first run served as a 5-epoch operational baseline to validate end-to-end pipeline correctness, achieving a best validation Dice of 0.310. The primary model was trained for 100 epochs under identical hyperparameters, reaching a final training Dice of 0.9504, training IoU of 0.9063, and a best validation Dice of 0.8507. Training convergence for both runs is detailed in Section IV-B.

B. Implementation Details

The model was implemented in PyTorch with mixed-precision training disabled, keeping the pipeline reproducible

on commodity hardware without distributed training requirements. The encoder applies four successive DoubleConv blocks with max-pooling for downsampling, producing feature maps at strides 1, 2, 4, and 8 relative to the input resolution. Each DoubleConv block consists of two 3×3 convolutions followed by batch normalization and ReLU activation. The decoder mirrors this structure with bilinear upsampling followed by convolutional refinement at each scale. Skip connections carry encoder feature maps directly to the corresponding decoder stage, preserving fine spatial detail lost during downsampling. The total number of trainable parameters is approximately 7.8 million. Checkpoints were saved at every epoch and the best validation Dice checkpoint was used for all reported evaluations. Fig. 6 illustrates the convergence behaviour across both training runs, confirming stable loss reduction and progressive metric improvement over 100 epochs.

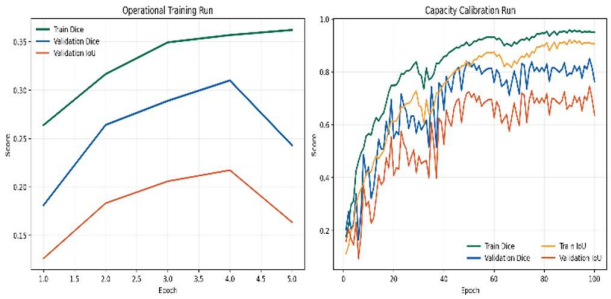


Fig 6. Training curves: 5-epoch operational baseline (left) and 100-epoch capacity calibration run (right). Primary model uses the 100-epoch checkpoint.

V. Results and Discussions

A. Quantitative Results

The trained Siamese U-Net was evaluated on the 18-tile validation split. Overall pixel accuracy reached 88.27% with an AUC of 0.8435. The macro-average Dice score is 0.4874 and macro-average IoU is 0.3811; these macro figures are depressed by the Destroyed class scoring 0.0 due to extreme pixel scarcity, and weighted averages better represent operational performance under this class imbalance. Per-class results are reported in Table II and visualized in Fig. 8.

TABLE I. Overall Performance Metrics on the 18-Tile Validation Split

Performance Metric	Result
Overall Accuracy	0.8827
Overall AUC	0.8435
Macro-avg Dice Coefficient	0.4874
Macro-avg IoU	0.3811
Best Validation Epoch	90
Train Dice (100-epoch run)	0.9504

TABLE II. Per-Class Dice and IoU Scores on the 18-Tile Validation Split

Class	Dice	IoU
Background	0.9428	0.8917
No Damage	0.6328	0.4628
Minor Damage	0.4734	0.3101
Major Damage	0.3881	0.2408
Destroyed	0.0000	0.0000

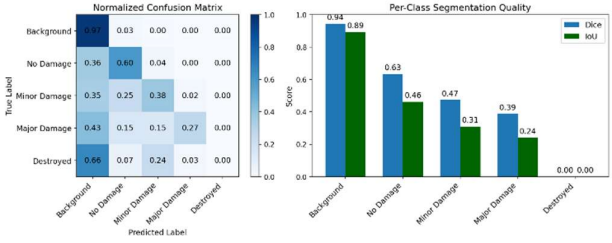


Fig 7. Normalized confusion matrix (left) and per-class Dice/IoU scores (right) on the 18-tile validation split.

B. Qualitative Results

Three representative validation tiles are shown in Fig. 7. AOI3-tile_5-5 shows strong structural alignment between prediction and ground truth for no-damage and minor-damage classes. AOI1-tile_1-3 is a heavily flooded scene with partial cloud occlusion where the model over-predicts major damage in the flooded corridor. AOI1-tile_5-5 shows good building-level localization but elevated false positives for the destroyed class in dense residential areas.



Fig 8. Qualitative results on three validation tiles. Columns show post-disaster image, ground truth overlay, and predicted overlay. Color: green=no damage, yellow=minor, orange=major, red=destroyed.

C. Discussion

The proposed model demonstrates strong and consistent performance on the Background and No Damage classes, achieving Dice scores of 0.9428 and 0.6328 respectively. This indicates that the per-scale temporal difference fusion strategy is effective in capturing meaningful structural changes between pre-disaster and post-disaster satellite imagery. The relatively high overall pixel accuracy of 88.27% and an AUC of 0.8435 further confirm that the model is capable of reliable large-scale damage identification, making it suitable for rapid assessment scenarios where quick triaging of affected regions is critical.

However, the results also reveal a clear limitation in handling rare and severe damage classes. The performance gradually declines from Minor Damage (0.4734) to Major Damage (0.3881), while the Destroyed class achieves a Dice score of 0.0000, primarily due to extreme class imbalance in the dataset. The confusion matrix further highlights a strong bias

toward predicting the Background class, where nearly 66% of Destroyed pixels are incorrectly labeled as Background, and around 43% of Major Damage regions are also misclassified as Background. This pattern suggests that the model is largely influenced by the dominant class distribution, causing it to prioritize common patterns over learning discriminative features for rare but critical damage categories.

Despite these limitations, the explainability analysis provides useful insights into the model's decision-making process. Grad-CAM visualizations show that the network consistently focuses on key structural elements such as buildings, road networks, and dense urban regions, which aligns well with expected damage-related cues in satellite imagery. Similarly, Score-CAM produces comparable attention maps without relying on gradient information, reinforcing the reliability of the highlighted regions. The agreement between both methods adds confidence that the model is attending to meaningful spatial regions rather than irrelevant background noise.Overall, while the model performs effectively for general damage detection and early-stage disaster assessment, improving sensitivity toward rare and severe damage classes remains an important direction for future work. Techniques such as advanced rebalancing strategies, focal loss optimization, and dataset expansion could help address the current bias and enhance robustness in real-world disaster scenarios.

VI. Conclusion

This study introduces a Siamese U-Net architecture with change-aware, per-scale temporal fusion for five-class building damage segmentation using xBD-style satellite imagery. To mitigate severe class imbalance, a combined Soft Dice loss and weighted Cross-Entropy loss is employed, resulting in an overall pixel accuracy of 88.27% and an AUC of 0.8435 on an 18-tile validation set. The integration of Grad-CAM and Score-CAM explainability modules generates spatially consistent saliency maps from the decoder pathway, enabling more interpretable and operational damage assessment. Future improvements will focus on enhancing rare-class detection through expanded datasets, the use of focal loss-based optimization, and a full evaluation of the proposed multimodal meteorological conditioning branch.

VII. Future Work

Future improvements will focus on four directions. First, scaling training to the full xBD benchmark dataset to improve rare-class coverage, particularly for the Destroyed category. Second, applying focal loss or targeted oversampling strategies to reduce the background bias observed in the confusion matrix. Third, activating and evaluating the multimodal meteorological conditioning pathway using rainfall, wind speed, elevation, and slope inputs at the bottleneck. Fourth, extending the dashboard to support batch inference across entire disaster zones and integration with geospatial mapping tools for field deployment.

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